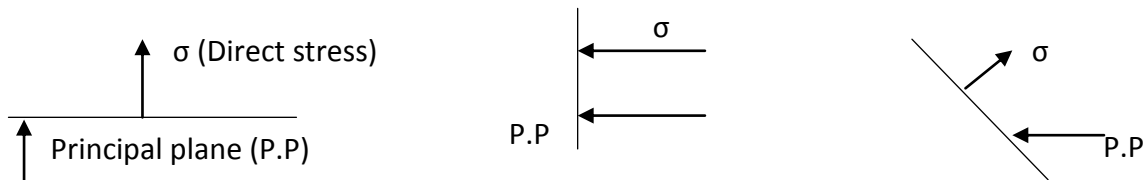


Principal Plane: -

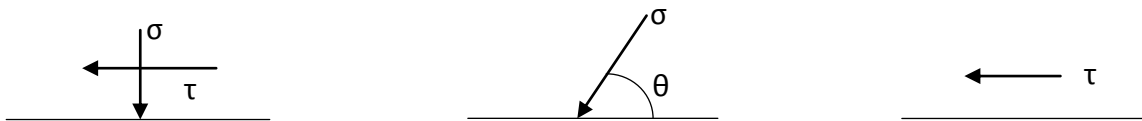
The plane on which only direct stress is acting is called Principal plane.



- on the Principal plane, shear stress is zero
- Principal plane may be vertical, horizontal or inclined. Planes shown in above figure are principal plane.

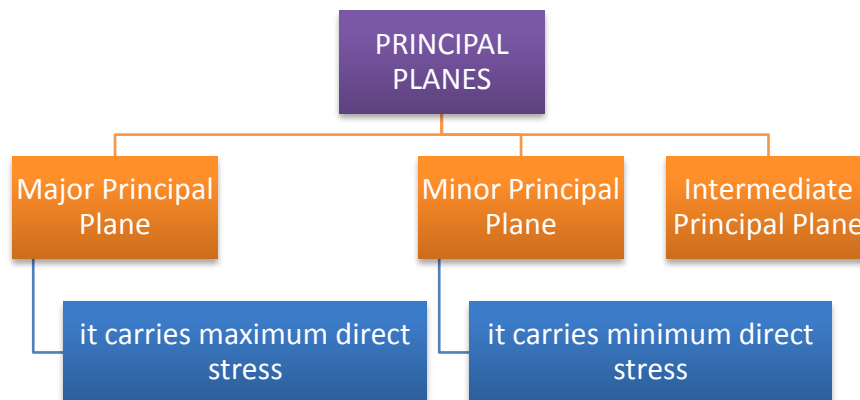
σ = (Direct stress), P.P = Principal plane.

The following planes are not principal plane as below figure.



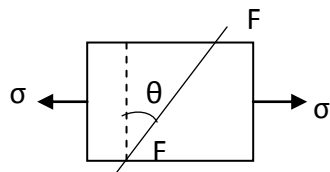
Principal stress: -

Maximum and Minimum values of stresses are acting normal to the principal plane is known as Principal Stresses.



Normal, Tangential & Resultant stress on an inclined plane.

- a) When body is Subjected to direct stresses in one plane.
(one Direct stress $\sigma_1 = \sigma$)

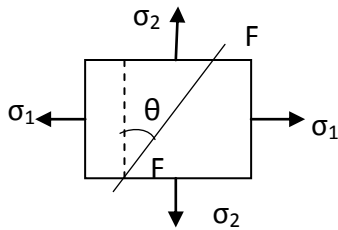


$$\sigma_n = \left(\frac{\sigma}{2}\right) - \left(\frac{\sigma}{2}\right) \cos 2\theta, \quad \sigma_t = \frac{\sigma}{2} \sin 2\theta$$

$$\sigma_r = \sqrt{\sigma_n^2 + \sigma_t^2}$$

$$\tau_{\max} = \frac{\sigma}{2}$$

- b) Body Subjected to direct stresses in two mutually perpendicular directions
(Direct stress σ_1 & σ_2)**

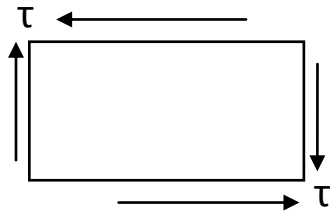


$$\sigma_n = \left(\frac{\sigma_1 + \sigma_2}{2} \right) - \left(\frac{\sigma_1 - \sigma_2}{2} \right) \cos 2\theta$$

$$\sigma_t = \left(\frac{\sigma_1 - \sigma_2}{2} \right) \sin 2\theta, \tau_{\max} = \frac{\sigma_1 - \sigma_2}{2}$$

$$\sigma_r = \sqrt{\sigma_n^2 + \sigma_t^2}$$

- c) Body subjected to a simple shear stress.**

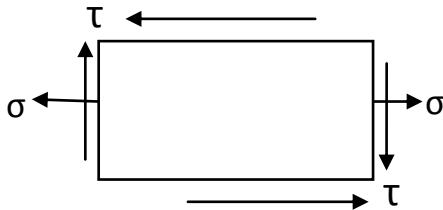


$$\sigma_n = -\tau \sin 2\theta$$

$$\sigma_t = \tau \cos 2\theta$$

$$\tau_{\max} = 0$$

- d) Body is Subjected to direct stress in one plane accompanied by a simple shear.**

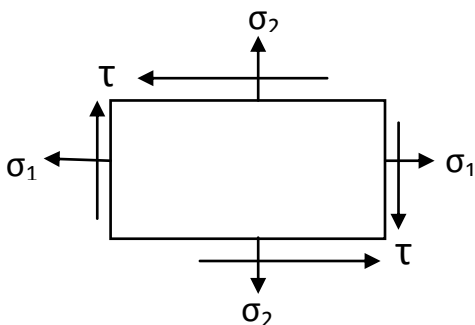


$$\sigma_n = \left(\frac{\sigma}{2} \right) - \left(\frac{\sigma}{2} \right) \cos 2\theta - \tau \sin 2\theta$$

$$\sigma_t = \frac{\sigma}{2} \sin 2\theta - \tau \cos 2\theta$$

$$\tau_{\max/\min} = \left(\frac{\sigma}{2} \right) + \sqrt{\left(\frac{\sigma}{2} \right)^2 + \tau^2}$$

- e) Body subjected to direct stresses in two mutually perpendicular direction accompanied by a simple stress (General equation)**

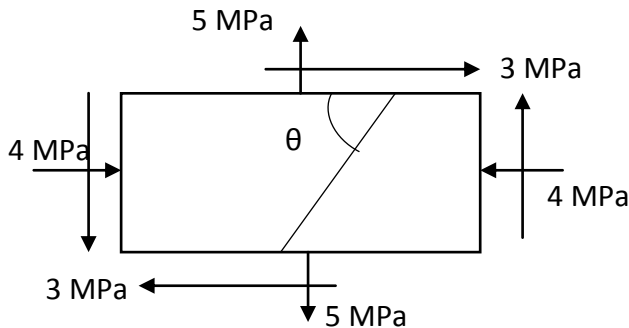


$$\sigma_n = \left(\frac{\sigma_1 + \sigma_2}{2} \right) + \left(\frac{\sigma_1 - \sigma_2}{2} \right) \cos 2\theta - \tau \sin 2\theta$$

$$\sigma_t = \left(\frac{\sigma_1 - \sigma_2}{2} \right) \sin 2\theta - \tau \cos 2\theta$$

$$\tau_{\max/\min} = \left(\frac{\sigma_1 + \sigma_2}{2} \right) + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2} \right)^2 + \tau^2}$$

Ex-1: An element is located by tensile stress 5 MPa & compressive stress 4 MPa in perpendicular directions, along with shear stress 3 MPa as shown in figure. Calculate normal, tangential & resultant stress on a plane making 30 angle in anti-clockwise direction with the plane carrying tensile stress.



$$\begin{aligned}\sigma_1 &= 5 \text{ N/mm}^2 \text{ (T)} \\ \sigma_2 &= -4 \text{ N/mm}^2 \text{ (C)} \\ \tau &= 3 \text{ N/mm}^2 \\ \theta &= -30^\circ \\ &\text{anti-clockwise to tensile stress.} \\ 2\theta &= -60^\circ\end{aligned}$$

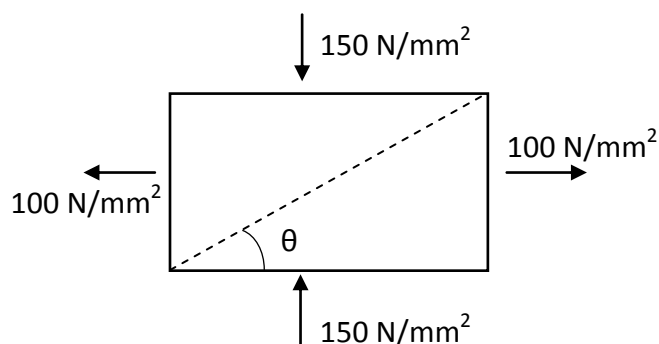
• **Normal Stress on plane: -**

$$\begin{aligned}\sigma_n &= \left(\frac{\sigma_1 + \sigma_2}{2}\right) + \left(\frac{\sigma_1 - \sigma_2}{2}\right) \cos 2\theta - \tau \sin 2\theta \\ &= \left(\frac{5-4}{2}\right) + \left(\frac{5-(-4)}{2}\right) \cos(-60^\circ) + 3 \sin(-60^\circ) \\ &= 0.15 \text{ N/mm}^2\end{aligned}$$

• **Tangential Stresses on plane.**

$$\begin{aligned}\sigma_t &= \left(\frac{\sigma_1 - \sigma_2}{2}\right) \sin 2\theta - \tau \cos 2\theta \\ &= \left(\frac{5-(-4)}{2}\right) \sin(-60^\circ) - 3 \cos(-60^\circ) \\ &= -5.4 \text{ N/mm}^2 \text{ (Compressive)}\end{aligned}$$

Ex-2: A direct stresses at a point in the strained material 150 N/mm² compressive & 100 N/mm² tensile. There is no shear stress. Find the normal & tangential stresses on the plane. Also find the resultant stress. Stress are acting at 30° degrees clockwise.



$$\begin{aligned}\sigma_1 &= -150 \text{ N/mm}^2 \text{ (C)} \\ \sigma_2 &= 100 \text{ N/mm}^2 \text{ (T)} \\ \tau &= 0 \\ \theta &= 30^\circ\end{aligned}$$

• **Normal Stress: -**

$$\begin{aligned}\sigma_n &= \left(\frac{\sigma_1 + \sigma_2}{2}\right) + \left(\frac{\sigma_1 - \sigma_2}{2}\right) \cos 2\theta - \tau \sin 2\theta \\ &= \left(\frac{-150 + 100}{2}\right) + \left(\frac{-150 + 100}{2}\right) \cos 60 + 0 \\ &= -87.5 \text{ N/mm}^2\end{aligned}$$

• **Tangential Stress: -**

$$\begin{aligned}\sigma_t &= \left(\frac{\sigma_1 - \sigma_2}{2}\right) \sin 2\theta - \tau \cos 2\theta \\ &= \left(\frac{-150 - 100}{2}\right) \sin 60 - 0 \\ &= -108.25 \text{ N/mm}^2\end{aligned}$$

• **Resultant Stress: -**

$$\begin{aligned}\sigma_r &= \sqrt{\sigma_n^2 + \sigma_t^2} \\ &= 139.194 \text{ N/mm}^2\end{aligned}$$

To find principal stresses & locate principal planes.

$$\sigma_{n1} = \left(\frac{\sigma_1 + \sigma_2}{2}\right) + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2}$$

σ_{n1} = Major Principal stress

$$\sigma_{n2} = \left(\frac{\sigma_1 + \sigma_2}{2}\right) - \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2}$$

σ_{n2} = Minor principal stress

$$\tau_{\max} = \left(\frac{\sigma_1 - \sigma_2}{2}\right)$$

$\tau_{\max}$ = maximum shear stress

$$\tan 2\alpha_1 = \left(\frac{2\tau}{\sigma_1 - \sigma_2}\right)$$

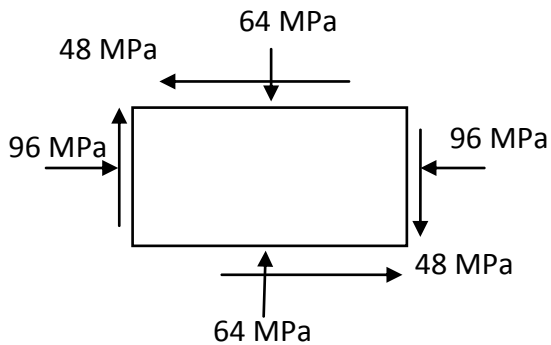
α_1 = angle of major principal plane with vertical plane

α_2 = angle of minor principal plane with vertical plane

$$\alpha_2 = \alpha_1 + 90^\circ$$

- Major principal plane and minor principal plane are always at right angle (90°).

Ex: -3 A point in two-dimensional stressed body is shown in figure. Determine the magnitudes & directions of principal stresses, using analytical method.



Here all stress produces compressive effect so as per sign conventional it produces negative stresses. So here we take compression as positive stress.

$$\sigma_1 = 96 \text{ MPa}$$

$$\sigma_2 = 64 \text{ MPa}$$

$$\tau = 48 \text{ MPa}$$

- Principal stresses**

$$\sigma_{n1} = \left(\frac{\sigma_1 + \sigma_2}{2} \right) + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2} \right)^2 + \tau^2}$$

$$\sigma_{n1} = 130.59 \text{ Mpa.}$$

$$\sigma_{n2} = \left(\frac{\sigma_1 + \sigma_2}{2} \right) - \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2} \right)^2 + \tau^2}$$

$$\sigma_{n2} = 29.41 \text{ MPa.}$$

- Angle of Principal plane.**

$$\tan 2\alpha_1 = \left(\frac{2\tau}{\sigma_1 - \sigma_2} \right)$$

$$= \left(\frac{2 \times 48}{96 - 64} \right)$$

$$\tan 2\alpha_1 = 3$$

$$2\alpha_1 = 71.56^\circ$$

$$\alpha_1 = 35.78^\circ$$

$$\alpha_2 = \alpha_1 + 90^\circ$$

$$\alpha_2 = 125.78^\circ$$